

Packaging, deploying, and running a Node application using Docker

Objective:

In this exercise, you will learn how to create a docker for a Node.js application.

Docker is a powerful tool that enables developers to package their applications and dependencies into a container, making it easier to deploy, manage, and scale their applications.

Although the following is a simple example, the same procedure can be applied to a more complex project.

Prerequisites

Before starting the exercise, you should install Node and Docker Desktop on your local PC.

Steps by step instructions

- Create a Node.js application: that prints "Hello World" in browser
- You will then create a Node app which serves some content.
- You will also create a file called **package.json** (using the **npm init** command) which defines all the project dependencies and sets other settings.
You will also experiment with installing a common module called "express"..

For this exercise, we will create a new directory called **docker_app** and create a file called **server.js** inside it which has a few lines of JavaScript for running a Node application.

```
mkdir docker_app
cd docker_app
npm init -y
code .
```

- Create a file called **server.js** and then copy the following code inside:

```
'use strict';

const express = require('express');

const PORT = 9000;
const HOST = '0.0.0.0';

const app = express();
app.get('/', (req, res) => {
  res.send('<h1>Hello World!</h1>');
});

app.listen(PORT, HOST, () => {
  console.log(`Running on http://${HOST}:${PORT}`);
});
```

- Change the contents of the **package.json** file to set the dependencies and other attributes.

```
{
  "name": "qa_node_app",
  "version": "1.0.0",
  "description": "A Node app using Docker",
  "author": "mike <mike@qa.com>",
  "main": "server.js",
  "scripts": {
    "start": "node server.js"
  },
  "dependencies": {
    "express": "^4.16.1"
  }
}
```

- Change the author's name and email
- Start a Terminal Window in vs-Code (press Ctrl-')
- Install all the dependencies by typing the following command in the Terminal window
npm install
- Start running your node app to see if it works by typing: **npm start**
- Open a browser and type: <http://localhost:9000>



You should see

- You can now stop the app running by pressing Ctrl-C in the Terminal window.

Create a Docker container for your app

Create a Dockerfile:

In this part, you will create a Dockerfile that will be used to build a Docker image for your Node.js application.

- Create a file called **Dockerfile** in the same folder as **server.js**
- Include the following instructions for basing the image on node version 16 and installing the files on the Docker.

```
FROM node:16

# Create app directory
WORKDIR /usr/src/app

# Install app dependencies
# An * is used to copy package.json AND package-lock.json
COPY package*.json ./

RUN npm install

# Bundle app source
COPY . .

EXPOSE 9000
CMD [ "node", "server.js" ] # use " not '
```

- Create a **.dockerignore** (dot dockerignore) file in the same directory as your Dockerfile with the following content:

```
node_modules
npm-debug.log
```

We will not copy the module files into the image because they can be created by the line **RUN npm install** in the Dockerfile.

Build the Docker image

- Open a terminal window, navigate to the root directory of your Node.js application
- Run the following command to build the Docker image with the tag of say **mike/qa-node-app**

docker build . -t mike/qa_node_app

Please note the dot **'.'** In the middle of the command.

You can change the tag but please make sure each letter of the tag is in lowercase

This command will use the Dockerfile to build a Docker image of your Node.js application and tag it with the name that you specify.

Run the Docker container:

- After the Docker image has been built, you can run a Docker container using the following command: **docker run -p 5000:9000 mike/qa_node_app**

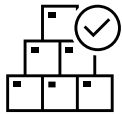
This command will start a new Docker container based on the node-docker-app image and map port **9000 in the container** to port **5000 on your local machine** (the host). So the format is **-p <host port>:<container port>**.

Test the Docker container:

- You should be able to access your Node.js application by opening a web browser and navigating to <http://localhost:5000>

Now remove your docker and its image

- Follow the instructions in the first Docker lab to remove the container and its image



Congratulations, you have successfully created a Docker to host a Node application.